

Graph Classes and Width Parameters in Lax

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This guide points to definitions in existing Lax submissions. Each marked passage opens the original concept and its Lean encoding beside the PDF. The source identifiers are also clickable in the PDF itself. The chapters are a reading order, not new definitions or claims of equivalence between formalizations. This submission has no concepts or proofs of its own.

Contents

1	Graph classes and closure	1
2	Sparse graph classes	1
2.1	Planar graph classes	1
2.2	Minors and treewidth	2
2.3	Admissibility and nowhere-dense classes	2
3	Dense graph classes	3
3.1	Cographs and width classes	3
3.2	Controlled density and class properties	3

1 Graph classes and closure

This opening section collects the vocabulary for talking about families of graphs and how they behave under standard operations.

Graph classes. The sparsity lectures represent a class by its finite simple graphs on canonical vertex sets, giving a common starting point for the later sparsity notions; see [lax-199508](#).

Isomorphism-closed graph classes. Another treatment makes closure under isomorphism explicit and provides union and intersection operations; see [lax-871432](#).

Closure properties. This concept groups closure under induced subgraphs, minors, contractions, disjoint-union summands, and disjoint union. In particular, induced-subgraph closure is the usual hereditary property; see [lax-871432](#).

Graph parameters. This concept gives a shared signature for natural-number graph parameters and records when two parameters bound one another up to functions; see [lax-153141](#).

2 Sparse graph classes

The examples below move from familiar planar families to containment and width notions, then to the order parameters and class properties used to describe nowhere-dense graphs.

2.1 Planar graph classes

The order follows the inclusions where they apply: paths and stars are subclasses of trees; trees, ladders, and maximal outerplanar graphs are outerplanar; and outerplanar graphs, series-parallel graphs, and triangulations are planar. The triangle family is included among the maximal outerplanar graphs.

Paths. These are the finite path graphs, the simplest connected examples in the list; every path is a tree. See [lax-68](#).

Stars. These trees have one centre joined to all other vertices; their leaves have no edges between them. See [lax-68](#).

Trees. A tree is connected and acyclic, so both paths and stars form subclasses of the tree family. See [lax-68](#).

Ladders. A ladder is a two-row grid, with two paths joined by corresponding rungs. See [lax-68](#).

Triangles. The concept singles out the three-vertex complete graph, a basic maximal outerplanar example. See [lax-68](#).

Maximal outerplanar graphs. These are outerplanar graphs to which no further edge can be added while preserving outerplanarity. See [lax-68](#).

Outerplanar graphs. Every vertex can be placed on the boundary of a crossing-free drawing; this family is contained in the planar graphs. See [lax-68](#).

Series-parallel graphs. These graphs are built from a terminal edge by series and parallel composition; the concept makes the terminal and support conditions explicit. See [lax-68](#).

Triangulations. A triangulation is a maximal planar supergraph on the same vertices, so it describes a way to complete a planar graph. See [lax-68](#).

Planar graphs. The concept uses a crossing-free straight-line drawing in the plane as its certificate. See [lax-68](#).

2.2 Minors and treewidth

Minor containment compares graphs by deletions and contractions; treewidth measures how closely a graph can be assembled from small overlapping bags. Grids and walls are standard planar examples that connect these viewpoints.

Graph minors. The branch-set formulation packages vertex deletion, edge deletion, and contraction into one containment relation. See [lax-68](#).

Topological minors. Here edges of the smaller graph are represented by internally disjoint paths, rather than arbitrary connected branch sets. See [lax-68](#).

Induced minors. This variant preserves both edges and non-edges between the retained branch sets, making it stricter than ordinary minor containment. See [lax-762056](#).

Grids and walls. The concept defines rectangular grids and their brick-wall subgraphs, recurring examples in planar structure theorems. See [lax-68](#).

Excluded induced grid minors. This class condition forbids an induced minor isomorphic to a square grid of the specified size. See [lax-762056](#).

Treewidth. Tree decompositions measure the largest bag needed to cover vertices and edges while keeping each vertex's bags connected. See [lax-228581](#).

2.3 Admissibility and nowhere-dense classes

These notions describe sparsity through bounded-radius paths, reachability orders, and the exclusion of dense shallow minors.

Admissibility. For an ordering, this parameter counts how many short paths from one vertex can reach earlier vertices while meeting only at their start. See [lax-199508](#).

Generalized coloring numbers. Weak and strong reachability count vertices accessible along short paths under a linear ordering; the coloring numbers minimize the worst such count over orderings. See [lax-199508](#).

Uniform quasi-wideness. Large vertex sets contain large distance-separated subsets after removing a bounded-size separator, uniformly over the class. See [lax-199508](#).

Shallow-minor density. This records edge density after bounded-depth contractions, at each fixed radius. See [lax-199508](#).

Shallow topological minors. This companion notion bounds the subdivision depth of the paths representing edges. See [lax-199508](#).

Nowhere-dense graph classes. At every fixed depth, some clique is excluded as a shallow minor of every member. See [lax-199508](#).

Almost-linear neighborhood complexity. This bounds the number of distinct neighborhood traces on any vertex set by a near-linear function of its size. See [lax-199508](#).

3 Dense graph classes

These concepts describe graph classes that can contain dense graphs, together with parameters used to organize them. The inclusion order is explicit for cographs and bounded cliquewidth; the other notions provide separate ways to control dense structure.

3.1 Cographs and width classes

Cographs. These are exactly the graphs of twin-width zero in the chosen formulation, a restricted dense-capable family. See [lax-214022](#).

Cliquewidth. A k -expression builds a graph using k labels; bounded cliquewidth therefore organizes classes by the number of labels needed. Cographs have cliquewidth at most two. See [lax-271696](#).

Merge-width. This parameter measures merge sequences using the radius- r neighborhoods in their partially resolved graphs. See [lax-9](#).

Twin-width. The partition-sequence parameter counts red neighbors created as vertex parts are merged, and applies to both sparse and dense families. See [lax-228581](#).

Mixed minor number. This parameter looks for fully mixed blocks in ordered adjacency matrices, giving another measure comparable to twin-width. See [lax-153141](#).

3.2 Controlled density and class properties

Weakly sparse classes. Such a class excludes one fixed complete bipartite graph as a subgraph; this still allows edge counts that grow superlinearly. See [lax-710763](#).

Monadically dependent classes. The concept characterizes classes that cannot produce every graph by first-order transductions with vertex colorings. See [lax-710763](#).

χ -boundedness. A class is controlled when chromatic number is bounded as a function of clique number, even if the class itself is dense. See [lax-9](#).

Linear neighborhood complexity. This counts distinct neighborhood traces and asks for a linear bound in the size of the tested vertex set. See [lax-9](#).

Feedback vertex number. This measures how many vertices must be removed to leave a forest; see [lax-379983](#).

Feedback edge number. Its edge-deletion counterpart removes edges until the remaining graph is acyclic; see [lax-379983](#).